



The 66th ASH Annual Meeting Abstracts

POSTER ABSTRACTS

722.ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD AND IMMUNE RECONSTITUTION

The Dominant Effect of the Gut Microbiome on GvHD in Mouse Models of Allo-BMTShayna Jankowski, BS¹, Shuhua Wang, MD¹, Yiwen Helen Li, MD², Cynthia R. Giver, PhD³, Edmund K. Waller, MDPhD^{1,4}¹Winship Cancer Institute, Emory University, Atlanta, GA²Emory University School of Medicine, Atlanta, GA³Emory University Winship Cancer Institute, Atlanta, GA⁴Department of Hematology and Medical Oncology, Emory University School of Medicine, Atlanta, GA

Introduction: The interaction between commensal microbiota and the mammalian intestinal immune system is crucial for health and disease, regulating the maturation of the host immune system and shaping immune responses in the gut (Zheng, Cell Research, 2020. 30: 492-506). The effects of gut microbes and their metabolites are key factors in regulating the development of acute intestinal graft-versus-host disease (aGVHD), a major factor in limiting the success of allogeneic bone marrow transplants (allo-BMT). Deconvoluting the microbiome in transplant recipients can predict GvHD severity and identify beneficial bacterial candidates. Using genetically identical mice with different gut microbiomes from different vendors, Jackson (JAX) and Taconic (TAC), allows for a direct comparison of the microbiome's effect on the incidence and severity of aGVHD in a mouse model. TAC mice harbor Segmented Filamentous Bacteria (SFB), which induce Th17 cell differentiation in the lamina propria, skewing the mucosal effector T cell balance (Ivanov, Mucosal Immunol. 2010 May; 3(3): 209-212). In contrast, the gut microbiome of JAX mice lacks SFB. We hypothesized that whole-genomic sequencing analysis of stool samples from TAC and JAX mice in an allo-BMT model would identify the bacterial species associated with more (or less) severe aGVHD.

Methods: C57BL/6 mice were purchased from JAX and TAC Laboratories. Vendor-driven microbiota diversity was preserved or enriched by separate housing or a co-housing approach (3 mice from vendor 1 mixed with 2 mice from vendor 2) for 2 weeks. Non-cohoused mice and co-housed mice were lethally irradiated (9 Gy) then transplanted with 5 x 10E6 T cell-depleted bone marrow (TCDBM) cells with or without 0.5 x 10E6 T cells from MHC-mismatched FVB/FVB-luc+ mice and monitored for survival. Stool samples were collected prior to co-housing, after co-housing, and at serial time points post-transplant. Microbial diversity was analyzed with CosmosID following shallow shotgun sequencing.

Results: Our study yielded a significant finding of higher GvHD-mortality rate ($p < 0.04$) in non-cohoused TAC mice, with 50% mortality over 60 days post-allo-BMT, compared to 10% mortality in JAX mice. Taxonomic sequencing analysis revealed greater microbial diversity in the gut of TAC mice relative to JAX, with a higher Shannon index for alpha-diversity ($p = 0.002$). The *Bacillota* to *Bacteroidota* ratio was significantly higher ($p = 0.007$) in TAC mice compared to JAX. TAC mice had more Th17 and Th1-inducing bacterial strains than JAX mice, such as *Prevotella*, *L. reuteri*, *M. schaedleri*, and *B. fragilis*. In contrast, TAC mice had fewer *Muribaculaceae* species and *Akkermansia muciniphila*. Following the co-housing of JAX and TAC mice, the microbiota diversity of JAX mice shifted to a TAC phenotype as determined by the Jaccard dissimilarity matrix. JAX mice co-housed with TAC mice were more vulnerable to GvHD, with 40% mortality compared to non-cohoused JAX mice ($p < 0.04$). Co-housed JAX mice that developed lethal aGVHD trended towards higher pre-transplant ratios of stool *Bacillota* to *Bacteroidota* compared with co-housed JAX mice that did not develop lethal aGVHD ($p < 0.10$).

Conclusion: The results indicate that murine GvHD survival is influenced by the ratio of *Bacillota* to *Bacteroidota* in the intestinal microbiome. Shotgun sequencing indicated an association of *Prevotella*, *L. reuteri*, *M. schaedleri*, and *B. fragilis* species with more severe GvHD. Profiling of metabolites involved in metabolic and catabolic pathways of the TAC and JAX microbiome will be analyzed in parallel with human plasma samples from BMT CTN studies of allogeneic transplants to identify metabolites and associated bacterial species that might provide clinical benefits in reducing aGVHD in transplant patients.

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